

**SAVEETHA INSTITUTE OF MEDICAL AND
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School of Engineering

Department of Artificial Intelligence & Machine Learning

CO2-Assessment Tool 2- Scenario Based

Submitted partial fulfilment of the requirements for

MLA0406 – Deep Learning

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Scenario-Based Assessment – Activation Functions Scenario

A college wants to develop a Deep Learning model to classify students into “Pass” and “Fail” based on their internal marks, attendance, and assignment marks.

The neural network has:

- Input layer: 3 neurons
- Hidden layer: 4 neurons
- Output layer: 1 neuron

During testing, the following problems are observed

1. When Sigmoid is used in the hidden layer, learning becomes very slow.
2. When ReLU is used, some neurons always produce an output of zero.
3. The output layer must produce a value between 0 and 1 representing the probability that a student will pass.

a) Which activation function would you recommend for the output layer? Give a reason.

Answer: Sigmoid Activation Function

The **Sigmoid activation function** is recommended for the output layer because the given problem is a **binary classification problem**, where students are classified into two categories: **Pass and Fail**. The Sigmoid function converts the neuron's output into a value ranging from **0 to 1**, which can be interpreted as the probability of a student passing. If the output probability is close to 1, the student has a higher chance of passing, whereas a value close to 0 indicates a higher chance of failing. Therefore, Sigmoid is an appropriate choice for the output layer.

$$\text{Sigmoid}(x) = \frac{1}{1 + e^{-x}}$$

b) Which activation function would you recommend for the hidden layer to overcome the slow-learning problem associated with Sigmoid?

Answer: ReLU (Rectified Linear Unit)

The **ReLU activation function** is recommended for the hidden layer because it can overcome the slow-learning problem commonly associated with Sigmoid. Sigmoid can suffer from the **vanishing gradient problem**, particularly when its input values become very large or very small. This causes the gradients to become very small and slows down the learning process. ReLU is computationally simple and provides a stronger gradient for positive inputs, allowing the neural network to learn faster and more efficiently. Hence, ReLU is widely used in hidden layers of deep learning networks.

$$ReLU(x) = \max(0, x)$$

For positive inputs, ReLU returns the input value, while for negative inputs, it returns zero.

c) What is the problem called when ReLU neurons continuously produce zero?

Answer: Dying ReLU Problem

When ReLU neurons continuously produce an output of zero, the problem is known as the **Dying ReLU problem**. ReLU produces zero whenever the input value is negative. If a neuron continuously receives negative inputs, its output remains zero and its gradient also becomes zero. As a result, the neuron may stop updating its weights during training and becomes ineffective in the neural network. This reduces the learning capability of the model and can negatively affect its performance. Therefore, continuously inactive ReLU neurons are referred to as **dead neurons**, and the condition is called the Dying ReLU problem.

d) Suggest one activation function that can overcome this problem.

Answer: Leaky ReLU

The **Leaky ReLU activation function** can be used to overcome the Dying ReLU problem. Unlike standard ReLU, Leaky ReLU does not completely eliminate negative inputs. Instead, it allows a small negative value to pass through the neuron. This ensures that a small gradient is available even when the input is negative, allowing the neuron to continue learning during training. Therefore, Leaky ReLU reduces the possibility of neurons becoming permanently inactive.

$$LeakyReLU(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x \leq 0 \end{cases}$$

Here, α is a small positive value, commonly taken as **0.01**.

e) Calculate its output using ReLU, Leaky ReLU and Sigmoid.

1. ReLU

$$ReLU(x) = \max(0, x)$$

If the input is positive, ReLU returns the same input value. If the input is negative, the output becomes zero.

2. Leaky ReLU

$$LeakyReLU(x) = \begin{cases} x, & x > 0 \\ 0.01x, & x \leq 0 \end{cases}$$

For positive inputs, Leaky ReLU behaves like ReLU. For negative inputs, it produces a small negative value instead of zero.

3. Sigmoid

$$Sigmoid(x) = \frac{1}{1 + e^{-x}}$$

Sigmoid converts any input into a value between **0 and 1**, making it useful for probability-based outputs.

ReLU:

$$ReLU(-2) = 0$$

Leaky ReLU:

$$LeakyReLU(-2) = 0.01(-2) = -0.02$$

Sigmoid:

$$Sigmoid(-2) = \frac{1}{1 + e^2} \approx 0.1192$$

So, the table becomes:

Activation Function	Input	Output
ReLU	-2	0
Leaky ReLU	-2	-0.02
Sigmoid	-2	0.1192

f) Based on the calculated values, explain why Leaky ReLU can be useful compared with ReLU for negative inputs.

Answer:

Leaky ReLU is useful compared with ReLU when the input is negative because **ReLU produces exactly zero for all negative inputs**. If a neuron repeatedly receives negative inputs, it may continuously produce zero and its gradient can become zero. This can cause the neuron to stop learning, resulting in the **Dying ReLU problem**.

Leaky ReLU solves this limitation by allowing a small negative value to pass through. For example, if the input is **-5** and $\alpha = 0.01$, ReLU produces **0**, whereas Leaky ReLU produces **-0.05**. This small negative output allows a gradient to remain available during training. Therefore, Leaky ReLU helps keep neurons active and improves learning for negative input values. It is particularly useful when preventing inactive neurons is important.